Week 4 Decision Gate: *King Octopus and the servants*

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Source and Problem Statement

The following puzzle is from Puzzle Prime:

King Octopus and His Servants

King Octopus has servants with 6, 7, or 8 legs. The servants with 7 legs always lie, and the servants with 6 or 8 legs always tell the truth. One day four of the king's servants had the following conversation:

> "We together have 28 legs," said the first octopus. "We together have 27 legs," said the second octopus. "We together have 26 legs," said the third octopus. "We together have 25 legs," said the fourth octopus.

Question: Which of the four servants told the truth? *Source:* Puzzle Prime, "King Octopus and His Servants."

Rules / Assumptions

Before solving the puzzle, we translate its natural-language statements into a precise mathematical model. The following assumptions describe the model used in this solution.

1. There are exactly four servants in the conversation.
2. Each servant has exactly one of the following numbers of legs:

$$6, \qquad 7, \qquad \text{or} \qquad 8.$$

3. When a servant says, "We together have n legs," the word "we" refers to the same group of all four servants in the conversation. Therefore, every servant is making a claim about the same total number of legs.
4. A servant with 7 legs always makes a false statement.
5. A servant with 6 or 8 legs always makes a true statement.
6. Each quoted sentence is treated as a complete mathematical statement. For example, "We together have 28 legs" means that the total is exactly 28, not approximately 28 and not at least 28.

7. The servants are counted once each, and "legs" has its ordinary meaning. We do not introduce missing legs, artificial legs, unmentioned servants, or other information not present in the problem.
8. The four statements refer to the same group at the same time. The total number of legs does not change during the conversation.

Let x_i be the number of legs belonging to servant i, where

$$x_i \in \{6, 7, 8\} \qquad \text{for } i \in \{1, 2, 3, 4\}.$$

Let the total number of legs be

$$T = x_1 + x_2 + x_3 + x_4.$$

The four servants are therefore asserting, in order,

$$T = 28, \qquad T = 27, \qquad T = 26, \qquad T = 25.$$

For each servant i, the truth value of that servant's statement is determined by the servant's number of legs:

$$x_i = 7 \quad \Longleftrightarrow \quad \text{servant } i\text{'s statement is false,}$$

and

$$x_i \in \{6, 8\} \quad \Longleftrightarrow \quad \text{servant } i\text{'s statement is true.}$$

These assumptions do not tell us in advance which servant is truthful. They only establish the rules that a proposed solution must satisfy.

Work

Let x_i be the number of legs belonging to servant i, and let

$$T = x_1 + x_2 + x_3 + x_4$$

be the total number of legs belonging to the four servants. The four statements are:

$$\begin{aligned} S_1 &: T = 28, \\ S_2 &: T = 27, \\ S_3 &: T = 26, \\ S_4 &: T = 25. \end{aligned}$$

A servant with 7 legs lies. A servant with 6 or 8 legs tells the truth.

Claim. *The second servant told the truth. The second servant has 6 legs, the other three servants each have 7 legs, and together the four servants have 27 legs.*

Proof. We first show that at least one servant must be telling the truth. Suppose, for the sake of contradiction, that all four servants are lying. Every liar must have 7 legs, so all four servants would have 7 legs. Their total would therefore be

$$T = 7 + 7 + 7 + 7 = 28.$$

But the first servant claimed that $T = 28$. The first servant's statement would then be true, contradicting our assumption that all four servants were lying. Therefore, at least one servant tells the truth. Next, observe that the four servants claim four different totals:

$$28, \qquad 27, \qquad 26, \qquad 25.$$

The group has only one actual total T, so no two of these different equalities can both be true. Therefore, at most one servant tells the truth. Since at least one servant tells the truth and at most one servant tells the truth, exactly one servant tells the truth. The other three servants lie. Each lying servant must have 7 legs, so the three liars contribute

$$3 \cdot 7 = 21$$

legs. The one truthful servant must have either 6 or 8 legs. If the truthful servant has 6 legs, then

$$T = 21 + 6 = 27.$$

If the truthful servant has 8 legs, then

$$T = 21 + 8 = 29.$$

A truthful servant's statement must equal the actual total. None of the four servants claimed that $T = 29$, but the second servant claimed that

$$T = 27.$$

Therefore, the truthful servant must have 6 legs, the actual total must be 27, and the truthful servant must be the second servant. Thus,

$$(x_1, x_2, x_3, x_4) = (7, 6, 7, 7),$$

and the second servant is the only servant who told the truth. $\square$

Check Your Answer

Either verify *every* proof step against a definition, or verify the counterexample really satisfies the hypothesis and violates the conclusion. For an argument-validity problem, exhibit premises that are all true while the conclusion is false (or argue that is impossible).

One-Sentence Summary

Exactly what you established — and, where it matters, distinguish "the conclusion happens to be true" from "the argument is valid."

Strongest disagreement

The best objection a sharp, skeptical reader would raise, stated honestly, and whether it survives.